

7. Write a method that accepts a `StringBuilder` object as an argument and converts all occurrences of the lowercase letter 't' in the object to uppercase.
8. Look at the following string:
`"cookies>milk>fudge:cake:ice cream"`
 Write code using the `String` class's `split` method that can be used to extract the following tokens from the string: `cookies`, `milk`, `fudge`, `cake`, and `ice cream`.
9. Assume that `d` is a `double` variable. Write an `if` statement that assigns `d` to the `int` variable `i` if the value in `d` is not larger than the maximum value for an `int`.
10. Write code that displays the contents of the `int` variable `i` in binary, hexadecimal, and octal.

Short Answer

1. Why should you use `StringBuilder` objects instead of `String` objects in a program that makes lots of changes to strings?
2. A program reads a string as input from the user for the purpose of tokenizing it. Why is it a good idea to trim the string before tokenizing it?
3. Each of the numeric wrapper classes has a static `toString` method. What do these methods do?
4. How can you determine the minimum and maximum values that may be stored in a variable of a given data type?

Programming Challenges

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1. Backward String

Write a method that accepts a `String` object as an argument and displays its contents backward. For instance, if the string argument is "gravity" the method should display -"ytivarg". Demonstrate the method in a program that asks the user to input a string and then passes it to the method.

2. Word Counter

Write a method that accepts a `String` object as an argument and returns the number of words it contains. For instance, if the argument is "Four score and seven years ago" the method should return the number 6. Demonstrate the method in a program that asks the user to input a string and then passes it to the method. The number of words in the string should be displayed on the screen.

3. Sentence Capitalizer

Write a method that accepts a `String` object as an argument and returns a copy of the string with the first character of each sentence capitalized. For instance, if the argument is "hello. my name is Joe. what is your name?" the method should return the string "Hello. My name is Joe. What is your name?" Demonstrate the method in a program that asks the user to



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input a string and then passes it to the method. The modified string should be displayed on the screen.

4. Vowels and Consonants

Write a class with a constructor that accepts a `String` object as its argument. The class should have a method that returns the number of vowels in the string, and another method that returns the number of consonants in the string. Demonstrate the class in a program that performs the following steps:

1. The user is asked to enter a string.
2. The program displays the following menu:
 - a. Count the number of vowels in the string
 - b. Count the number of consonants in the string
 - c. Count both the vowels and consonants in the string
 - d. Enter another string
 - e. Exit the program
3. The program performs the operation selected by the user and repeats until the user selects e, to exit the program.

5. Password Verifier

Imagine you are developing a software package for Amazon.com that requires users to enter their own passwords. Your software requires that users' passwords meet the following criteria:

- The password should be at least six characters long.
- The password should contain at least one uppercase and at least one lowercase letter.
- The password should have at least one digit.

Write a class that verifies that a password meets the stated criteria. Demonstrate the class in a program that allows the user to enter a password and then displays a message indicating whether it is valid or not.

6. Telemarketing Phone Number List

Write a program that has two parallel arrays of `String` objects. One of the arrays should hold people's names and the other should hold their phone numbers. Here are example contents of both arrays:

name Array Example Contents	phone Array Example Contents
"Harrison, Rose"	"555-2234"
"James, Jean"	"555-9098"
"Smith, William"	"555-1785"
"Smith, Brad"	"555-9224"

The program should ask the user to enter a name or the first few characters of a name to search for in the array. The program should display all of the names that match the user's input and their corresponding phone numbers. For example, if the user enters "Smith", the program should display the following names and phone numbers from the list:

```
Smith, William: 555-1785
Smith, Brad: 555-9224
```

7. Check Writer

Write a program that displays a simulated paycheck. The program should ask the user to enter the date, the payee's name, and the amount of the check. It should then display a simulated check with the dollar amount spelled out, as shown here:

	Date: 11/24/2012
Pay to the Order of: John Phillips	\$1920.85
One thousand nine hundred twenty and 85 cents	

8. Sum of Numbers in a String

Write a program that asks the user to enter a series of numbers separated by commas. Here is an example of valid input:

7,9,10,2,18,6

The program should calculate and display the sum of all the numbers.

9. Sum of Digits in a String

Write a program that asks the user to enter a series of single digit numbers with nothing separating them. The program should display the sum of all the single digit numbers in the string. For example, if the user enters 2514, the method should return 12, which is the sum of 2, 5, 1, and 4. The program should also display the highest and lowest digits in the string. (*Hint: Convert the string to an array.*)

10. Word Counter

Write a program that asks the user for the name of a file. The program should display the number of words that the file contains.

11. Sales Analysis

The file *SalesData.txt*, in this chapter's source code folder, contains the dollar amount of sales that a retail store made each day for a number of weeks. Each line in the file contains seven numbers, which are the sales numbers for one week. The numbers are separated by a comma. The following line is an example from the file:

2541.36,2965.88,1965.32,1845.23,7021.11,9652.74,1469.36

Write a program that opens the file and processes its contents. The program should display the following:

- The total sales for each week
- The average daily sales for each week
- The total sales for all of the weeks
- The average weekly sales
- The week number that had the highest amount of sales
- The week number that had the lowest amount of sales

12. Miscellaneous String Operations

Write a class with the following static methods:

- **WordCount**. This method should accept a reference to a `String` object as an argument and return the number of words contained in the object.
- **arrayToString**. This method accepts a `char` array as an argument and converts it to a `String` object. The method should return a reference to the `String` object.
- **mostFrequent**. This method accepts a reference to a `String` object as an argument and returns the character that occurs the most frequently in the object.
- **replaceSubstring**. This method accepts three references to `String` objects as arguments. Let's call them `string1`, `string2`, and `string3`. It searches `string1` for all occurrences of `string2`. When it finds an occurrence of `string2`, it replaces it with `string3`. For example, suppose the three arguments have the following values:

```
string1:  "the dog jumped over the fence"
string2:  "the"
string3:  "that"
```

With these three arguments, the method would return a reference to a `String` object with the value "that dog jumped over that fence".

Demonstrate each of these methods in a complete program.

13. Alphabetic Telephone Number Translator

Many companies use telephone numbers like 555-GET-FOOD so the number is easier for their customers to remember. On a standard telephone, the alphabetic letters are mapped to numbers in the following fashion:

```
A, B, and C = 2
D, E, and F = 3
G, H, and I = 4
J, K, and L = 5
M, N, and O = 6
P, Q, R, and S = 7
T, U, and V = 8
W, X, Y, and Z = 9
```

Write an application that asks the user to enter a 10-character telephone number in the format xxx-xxx-xxxx. The application should display the telephone number with any alphabetic characters that appeared in the original translated to their numeric equivalent. For example, if the user enters 555-GET-FOOD the application should display 555-438-3663.

14. Word Separator

Write a program that accepts as input a sentence in which all of the words are run together, but the first character of each word is uppercase. Convert the sentence to a string in which the words are separated by spaces and only the first word starts with an uppercase letter. For example, the string "StopAndSmellTheRoses." would be converted to "Stop and smell the roses."

15. Pig Latin

Write a program that reads a sentence as input and converts each word to “Pig Latin”. In one version of Pig Latin, you convert a word by removing the first letter, placing that letter at the end of the word, and then appending “ay” to the word. Here is an example:

English: I SLEPT MOST OF THE NIGHT
Pig Latin: IAY LEPTSAY OSTMAY FOAY HETAY IGHNTAY

16. Morse Code Converter

Morse code is a code where each letter of the English alphabet, each digit, and various punctuation characters are represented by a series of dots and dashes. Table 9-14 shows part of the code. Write a program that asks the user to enter a string, and then converts that string to Morse code. Use hyphens for dashes and periods for dots.

Table 9-14 Morse code

Character	Code	Character	Code	Character	Code	Character	Code
space	<i>space</i>	6	-....	G	--.	Q	--.-
comma	--..--	7	--...-	H		R	.-.
period	..-.-	8	---..	I	..	S	...
question mark	..-.-.	9	----.	J	.----	T	-
0	-----	A	.-	K	-.-	U	..-
1	.-----	B	-...-	L	.-..	V	...-
2	..----	C	-.-.	M	--	W	.--
3	...--	D	-..	N	-.	X	-.--
4	-	E	.	O	---	Y	-.--
5		F	..-.	P	.-.-.	Z	--..